



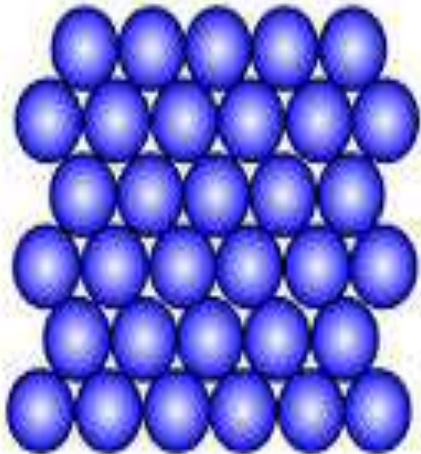
Introduction to solid state physics

- ❖ **Solid-state physics** is the study of rigid matter (solids) by examining how their large-scale properties result from their atomic-scale properties and interactions. It is the largest branch of condensed matter physics and provides the theoretical foundation for materials science
- ❖ **Quantum Mechanics:** Solid-state physics uses quantum mechanics to explain phenomena that classical physics could not, such as the heat capacity of metals and the existence of electrical insulators. Key quantum models include the free electron model and the nearly free electron model, which introduces electronic bands to explain conductivity.

❖ **Crystalline Solids:** Have a highly ordered, repeating, long-range pattern called a crystal lattice. Most of solid-state physics theory focuses on crystalline materials because their periodicity facilitates mathematical modeling.

Crystalline Solid

Particles



Example: Table Salt

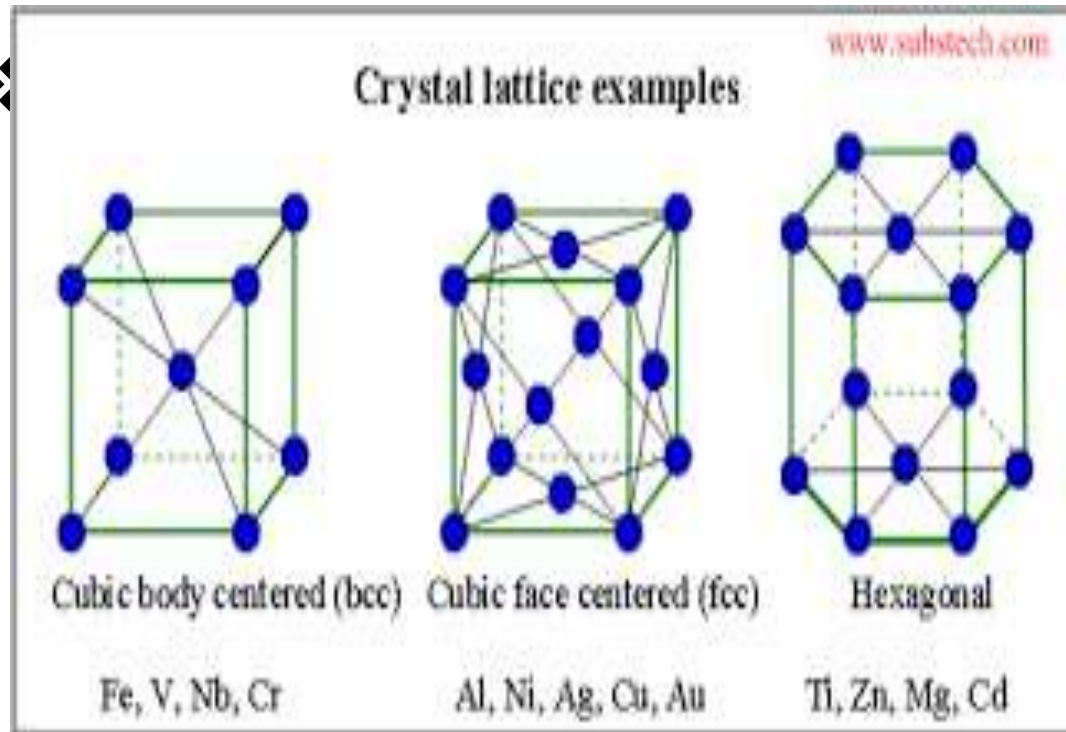


❖ **Amorphous solids** lack this long-range order, with atoms arranged randomly (e.g., glass, rubber).

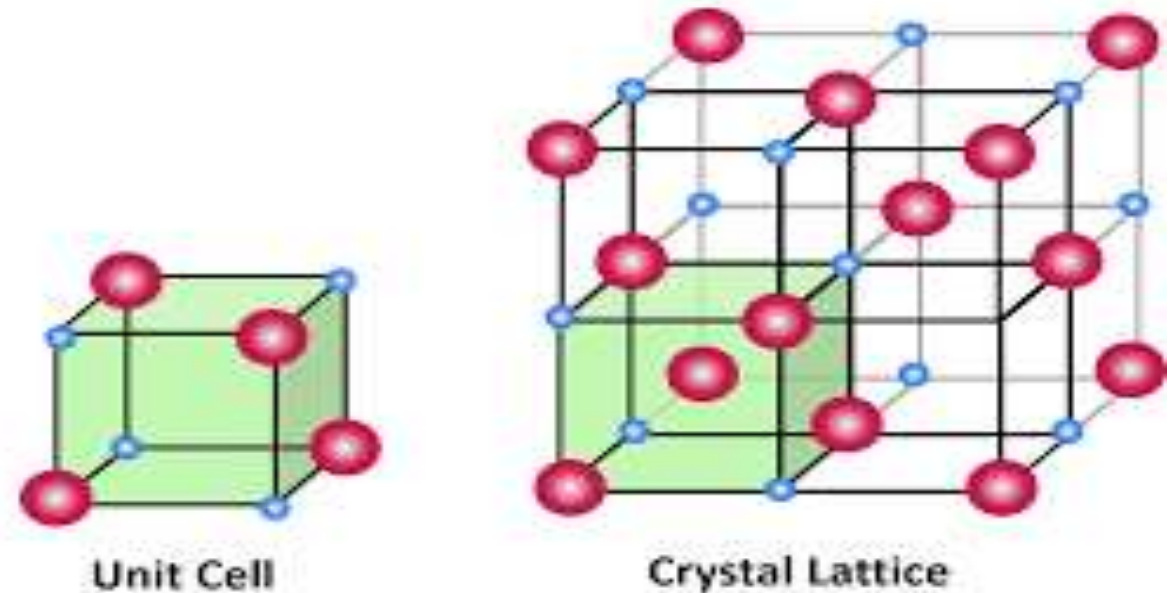


Amorphous

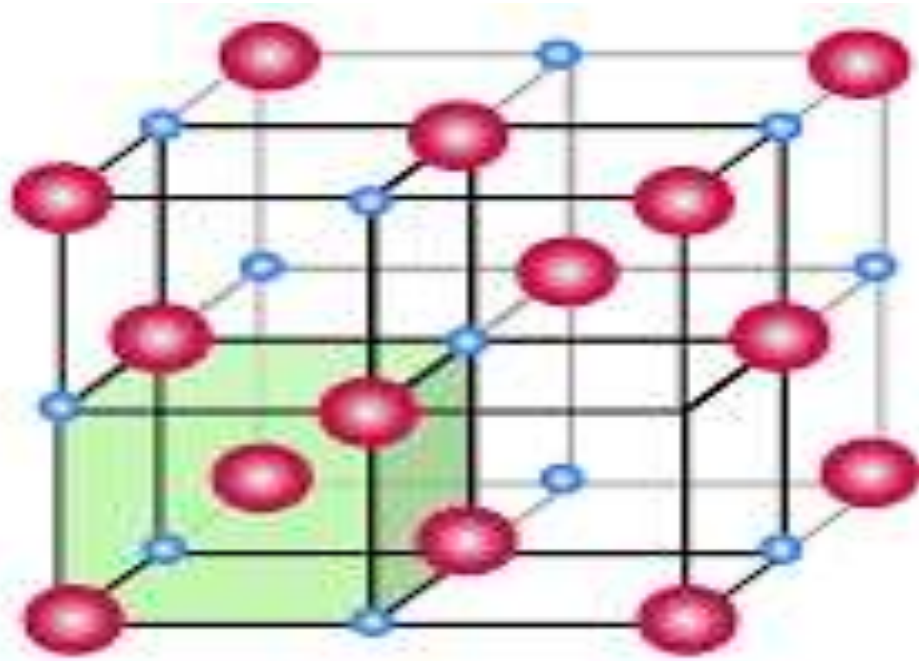
❖ **Crystal Structure:** This field extensively studies crystal structures, including fundamental lattice types (simple cubic, body-centered cubic, face-centered cubic), unit cells, and imperfections or defects within the structure, which significantly influence material properties.



Crystal Lattice and Unit Cell

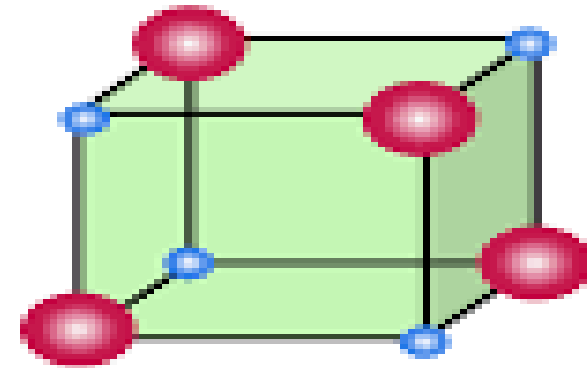


- ❖ A **crystal lattice** is a regular, three-dimensional arrangement of points representing the positions of atoms, ions, or molecules in a crystalline solid



Crystal Lattice

- ❖ A **unit cell** is the smallest repeating structural unit of this lattice, which, when repeated, generates the entire crystal structure



Unit Cell

Types of Materials

❖ **Conductor:** A **conductor** is a material that allows electric current or heat to flow through it easily because it has free-moving electrons or ions.



All Metals are conductors



Copper



Gold



Brass



Aluminum



Silver



Steel



Iron



Properties

- ❖ **High Conductivity / Low Resistance:** Conductors have high electrical and thermal conductivity and offer minimal resistance to the flow of charge or heat.
- ❖ **Free Electrons:** The presence of free-moving charge carriers (usually electrons in metals, ions in solutions) is the primary characteristic that enables conduction.
- ❖ **Zero Electric Field Inside:** In a static situation, the electric field inside a conductor is zero, and free charges reside only on its surface.

Applications

- ❖ Conductors are essential for modern technology and are found in numerous applications:
- ❖ **Electrical wiring** in homes and businesses.
- ❖ **Power transmission lines** that carry electricity from generating stations to consumers.
- ❖ **Electronic circuits** in devices like computers and cell phones.
- ❖ **Heating elements** in some appliances (where resistance is intentionally used to generate heat).
- ❖ **Automobile radiators** (as thermal conductors to dissipate heat from the engine)

❖ **Superconductors:** These are materials that, when cooled below a certain critical temperature, lose all electrical resistance, allowing current to flow with no energy loss.

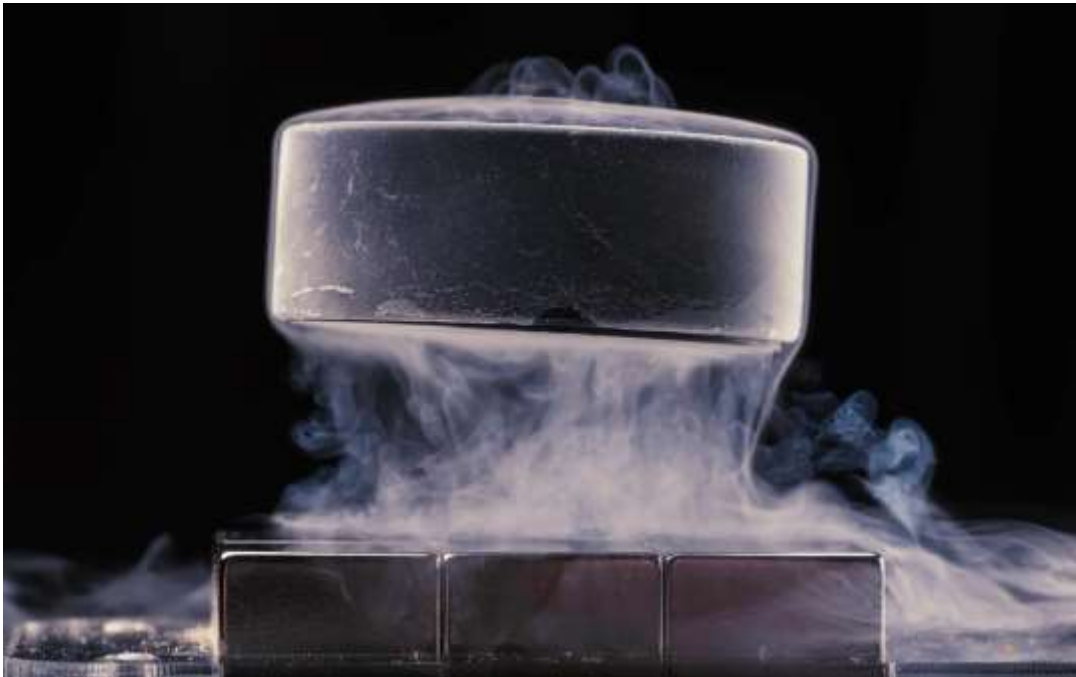
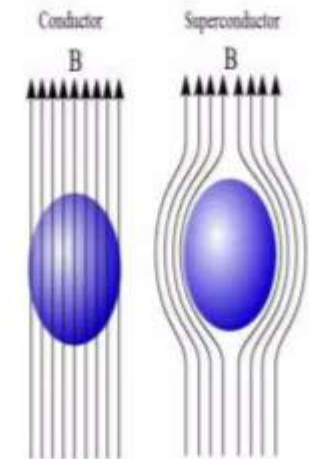
○ Definition & Basic Characteristics

✎ Superconductor: is an element, inter-metallic alloy, or compound that conducts electricity without resistance below a certain temperature, magnetic field, and applied current.

✎ Observed in 1911 by H.K. Onnes in liquid helium at temperature 4K.

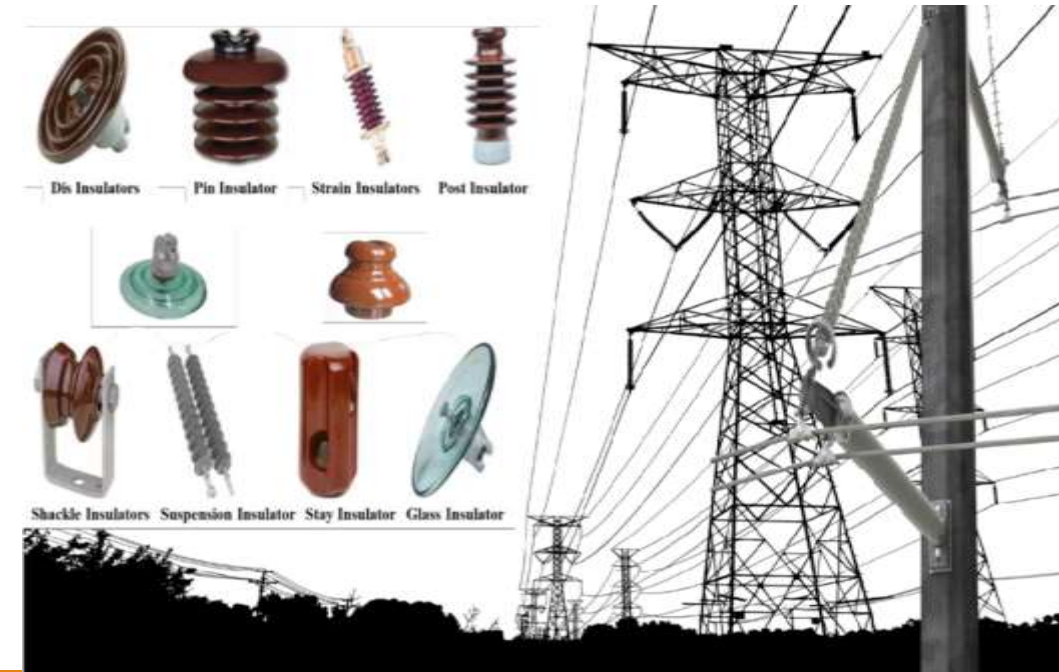
✎ In 1933 Meissner and Ochsenfeld discovered that a superconducting material repels a magnetic field.

This phenomenon is referred as Meissner effect.

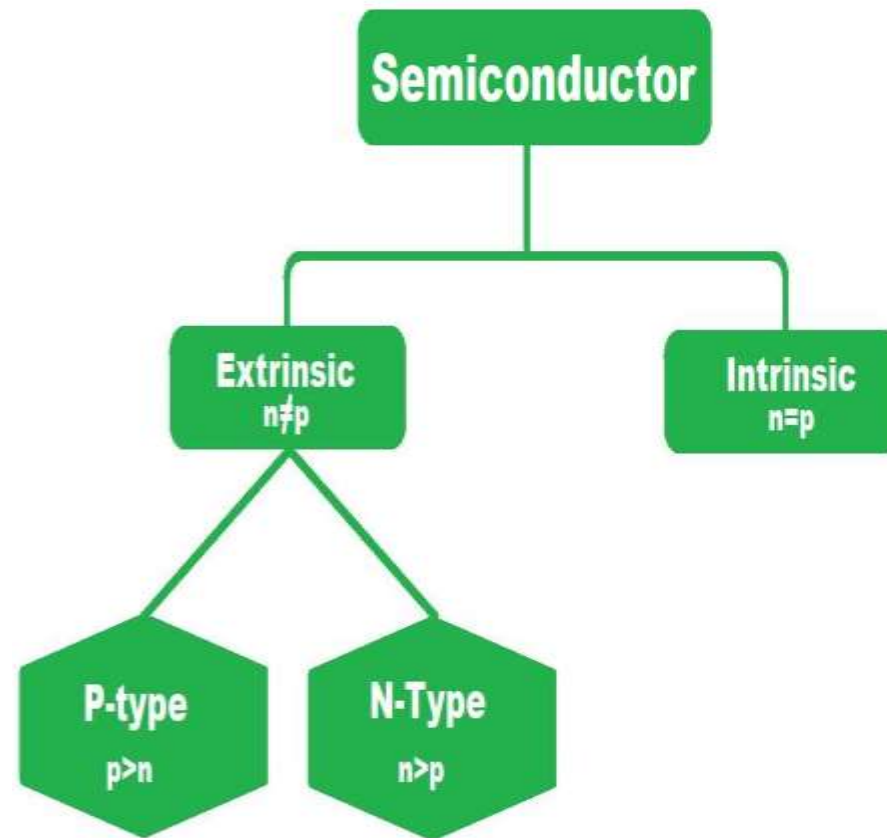


- ❖ **Applications** :Superconductors are used in a variety of advanced technologies due to their unique properties
- ❖ **Medical Equipment**: Powerful superconducting magnets are the core component of Magnetic Resonance Imaging (**MRI**) and Nuclear Magnetic Resonance (NMR) machines, which provide high-resolution images for medical diagnostics and chemical analysis.
- ❖ **Transportation: Maglev** (magnetic levitation) trains use superconducting magnets to levitate above the track, eliminating friction and allowing for very high speeds.
- ❖ **Scientific Research**: Superconducting magnets are essential for focusing and bending particle beams in high-energy **particle accelerators** like those at CERN. They are also used in experimental nuclear fusion reactors.
- ❖ **Energy and Power**: They have the potential for use in highly efficient power transmission lines (eliminating energy loss), powerful generators, and energy storage systems.
- ❖ **Computing**: Superconducting materials are used in highly sensitive magnetometers called Superconducting Quantum Interference Devices (SQUIDs) and are a key technology in developing **quantum computers**.

❖ **Insulators** are materials that block or significantly retard the flow of electrical current, heat, or sound. This property is a result of their atomic structure, where electrons are tightly bound to the atoms and cannot move freely, as they can in conductors.



❖ A **semiconductor** is a material with electrical conductivity between that of a good conductor (like copper) and an insulator (like rubber). Their unique property is that their conductivity can be precisely controlled, which is why they are the foundational material for virtually all modern electronics, including computer chips, transistors, and integrated circuits



➡ "n" stands for negative (free electrons)

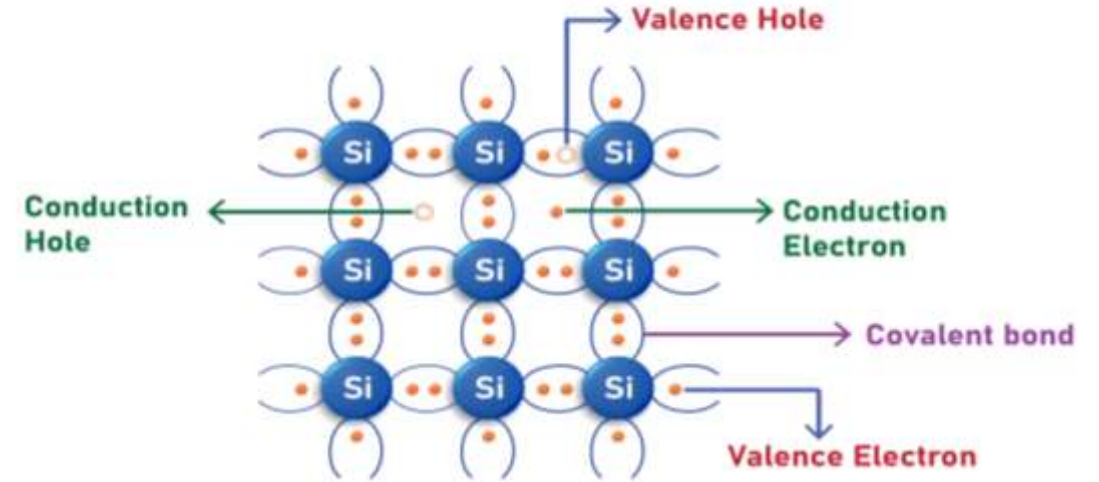
➡ "p" stands for positive (holes)

Semiconductor Components

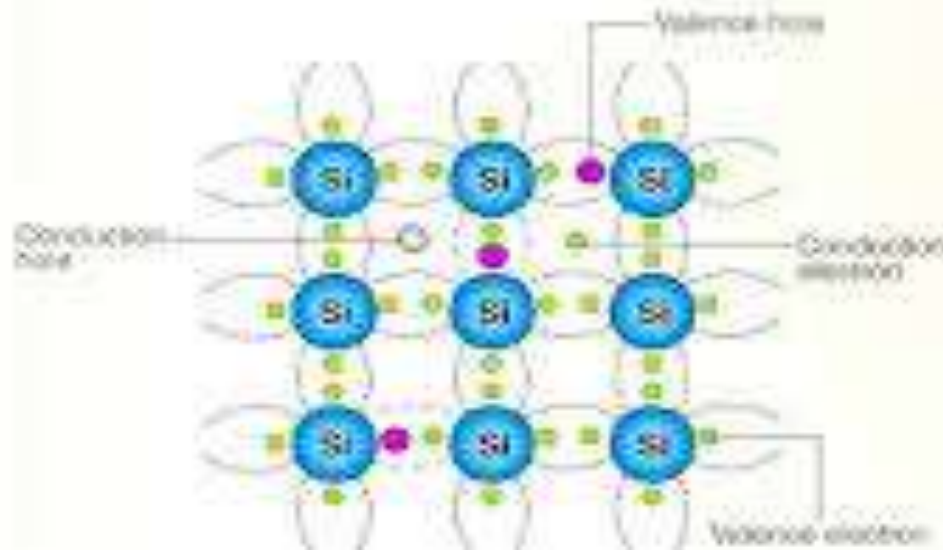


- ❖ **Intrinsic Semiconductors:** Pure semiconductors have equal numbers of electrons and holes, resulting in low conductivity.
- ❖ **Extrinsic Semiconductors:** Impure semiconductors, with added impurities, increase conductivity and are classified as N-type or P-type.

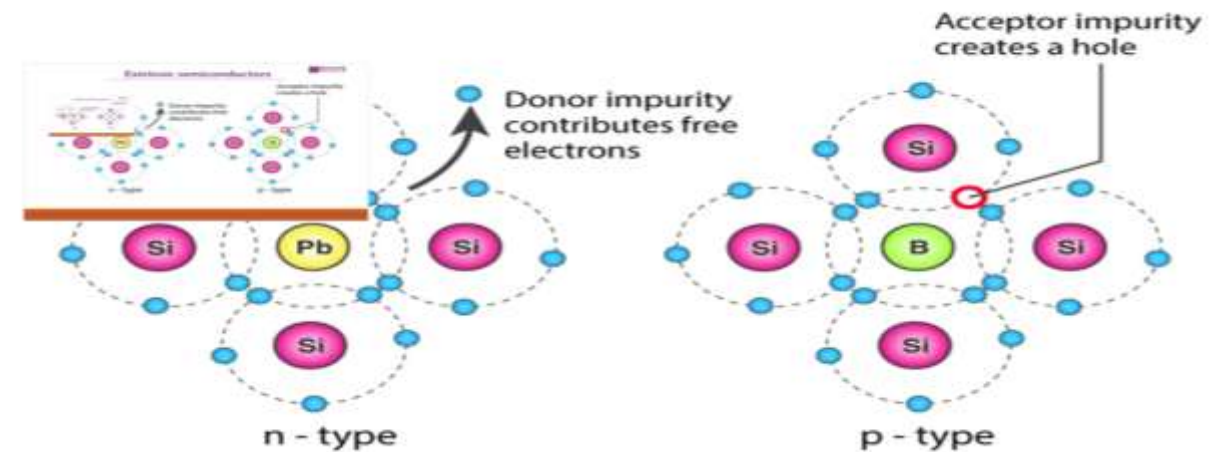
Intrinsic Semiconductor



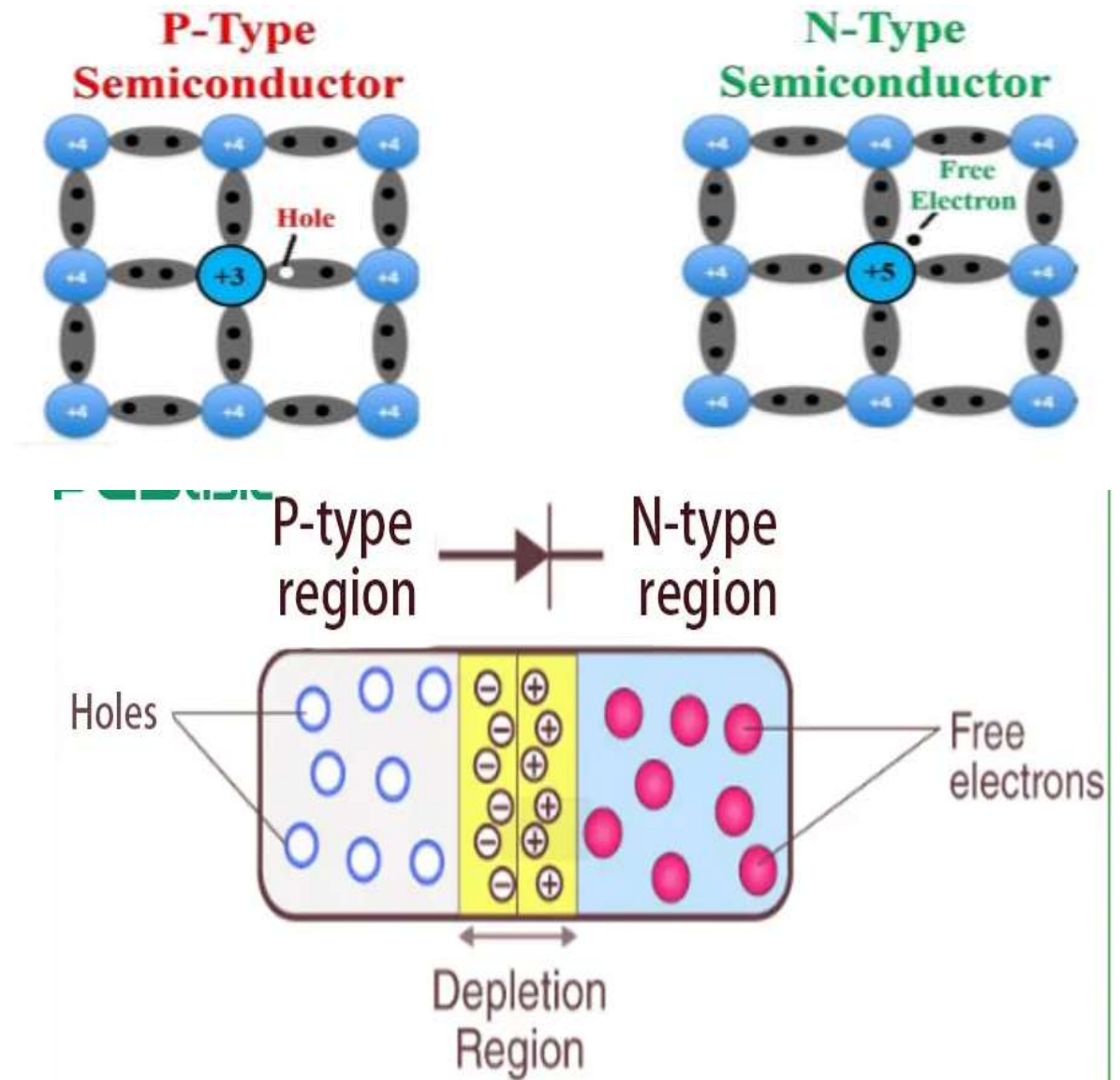
What is Extrinsic Semiconductor



Extrinsic semiconductors

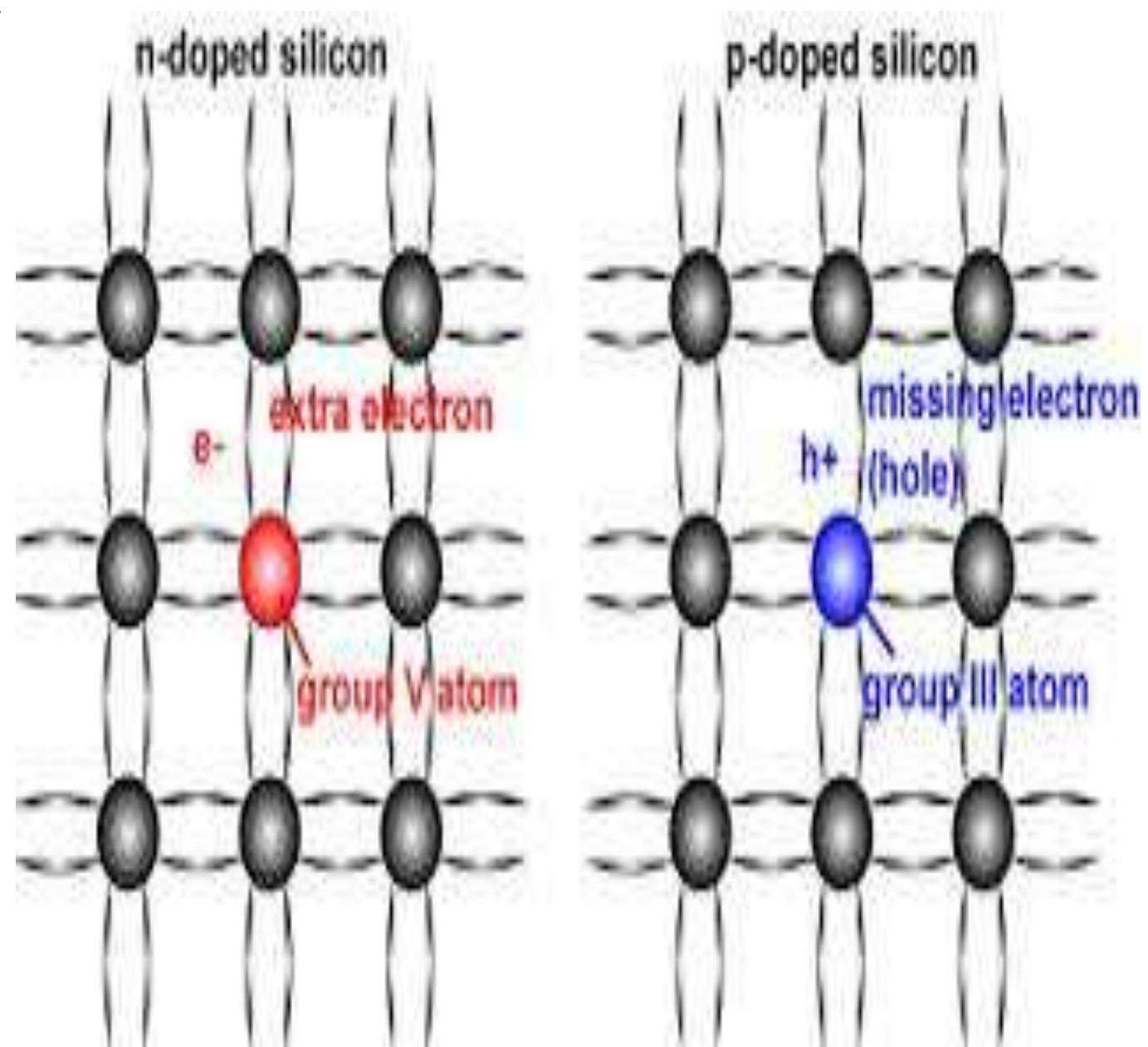


- ❖ **N-type and P-type** materials are extrinsic semiconductors created by a process called doping, where impurities are added to an intrinsic semiconductor (like silicon) to control its electrical conductivity.
- ❖ **N-type** semiconductors are created by doping a pure semiconductor with pentavalent impurities (Group V elements on the periodic table). These elements have five valence electrons (e.g., phosphorus, arsenic, or antimony).
- ❖ **P-type** semiconductors are created by doping with trivalent impurities (Group III elements). These elements have three valence electrons (e.g., boron, gallium, or indium).



❖ **Doping** is the intentional introduction of impurities (called **dopants**) into a pure (**intrinsic**) semiconductor material to modify its electrical properties, primarily its conductivity. The resulting material is an **extrinsic** semiconductor.

❖ **Purpose of Doping** Pure silicon or germanium has very low electrical conductivity at room temperature, making it unsuitable for most electronic devices. Doping dramatically increases the number of free charge carriers (electrons or "holes"), allowing controlled current flow in electronic components like diodes, transistors, and integrated circuits.



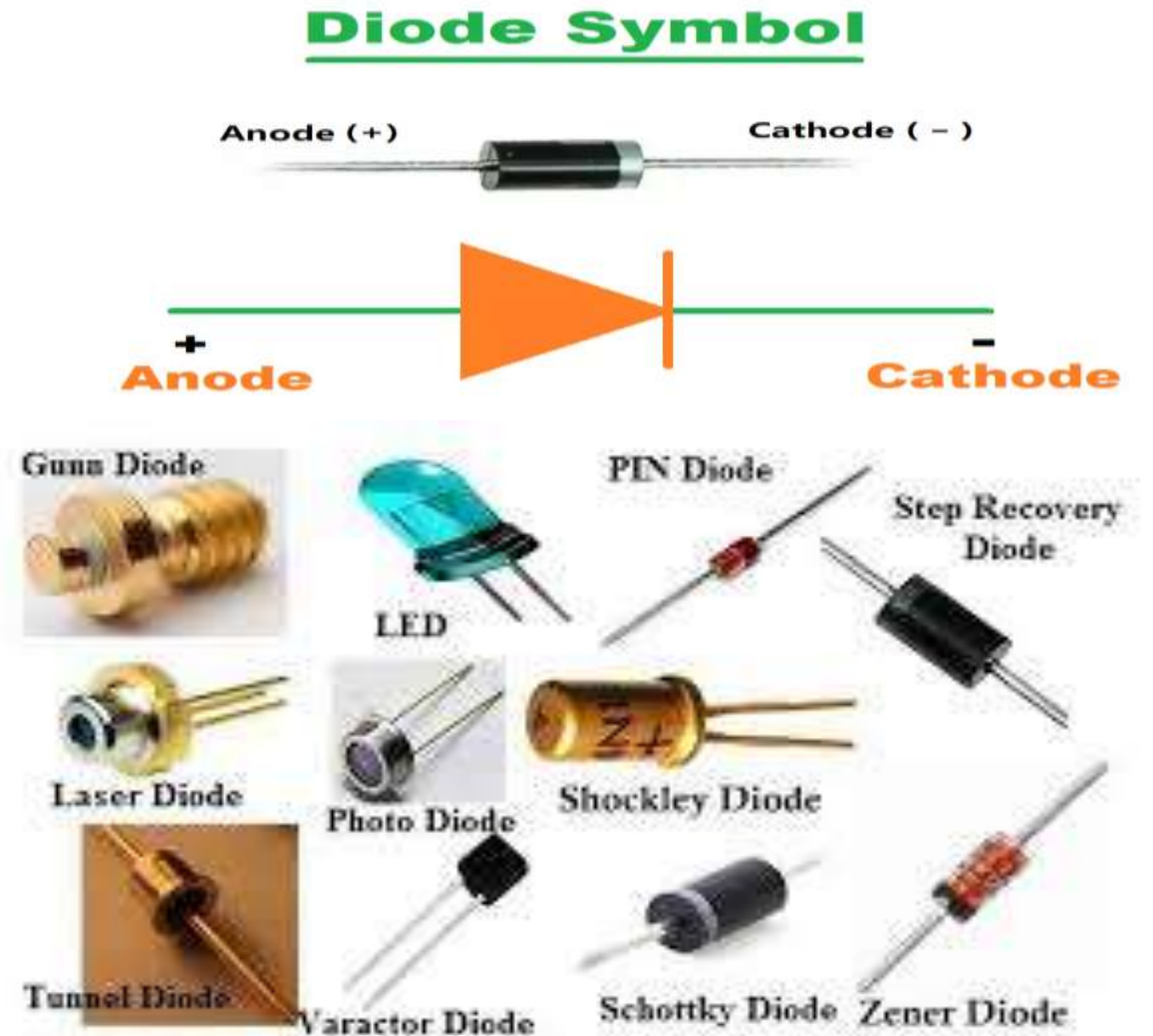
❖ Types of Doping

Doping creates two types of semiconductor material by introducing specific types of impurities

- ❖ **N-type (Negative-type):** Created by adding **donor** impurities from Group V of the periodic table (e.g., phosphorus, arsenic, antimony) to a Group IV semiconductor like silicon. These dopants have five valence electrons; four form covalent bonds with the surrounding silicon atoms, and the fifth becomes a free electron, which is the majority charge carrier.
- ❖ **P-type (Positive-type):** Created by adding **acceptor** impurities from Group III of the periodic table (e.g., boron, gallium, indium) to a Group IV semiconductor. These dopants have three valence electrons, resulting in a "hole" (a missing electron) in the crystal lattice. These holes behave as positive charge carriers, as electrons from neighboring atoms can move to fill the hole, creating a new hole in their previous location

❖ A **diode** is a two-terminal electronic component that acts as a **one-way switch** for electric current, allowing flow in one direction (forward bias) and blocking it in the opposite direction (reverse bias). This unidirectional property is essential in modern electronics

❖ Diodes are typically made of semiconductor materials like silicon or germanium, forming a **p-n junction**. A diode has a positive terminal, the **anode** (p-type), and a negative terminal, the **cathode** (n-type).



Diode Examples in Real Life

LED Indicators: Used in electronic devices like TVs, computers, and appliances to show power status.

Solar Panels: Photodiodes convert sunlight into electrical energy.

Smartphones: Zener diodes provide voltage regulation to protect circuits.

Car Alternators: Rectifier diodes convert AC to DC for the vehicle's electrical system.

Power Adapters: Rectifier diodes convert AC to DC for charging electronic devices.

Light Fixtures: LEDs are used for energy-efficient lighting in homes and offices.

Microwave Ovens: Diodes help generate and regulate microwave frequencies.

Television Remote Controls: Infrared LEDs transmit signals to the TV.

Laser Printers: Laser diodes are used for precise printing.

Radio Transmitters: Varactor diodes are used for frequency tuning.

Computer Power Supplies: Schottky diodes help in fast switching and efficient power conversion.

Burglar Alarms: Photodiodes detect changes in light levels to trigger alarms.

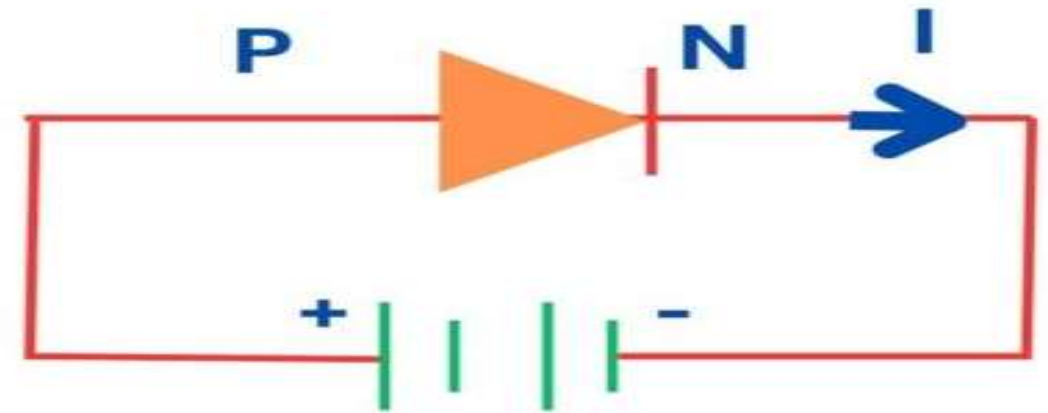
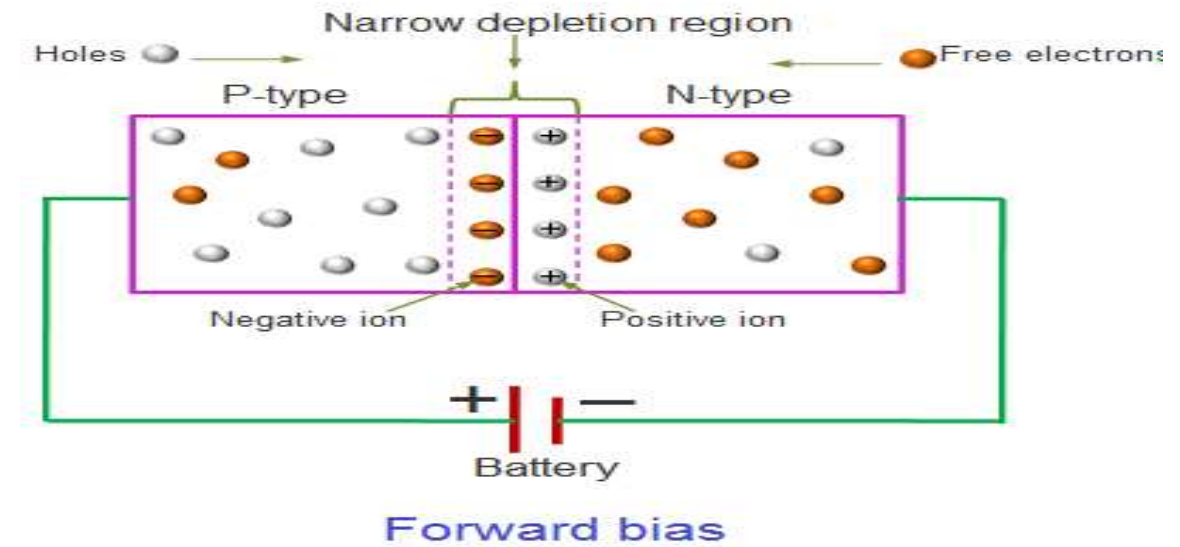
Voltage Surge Protectors: TVS diodes protect electronic devices from voltage spikes.

Fiber Optic Communications: Laser diodes are used to transmit data through optical fibers.

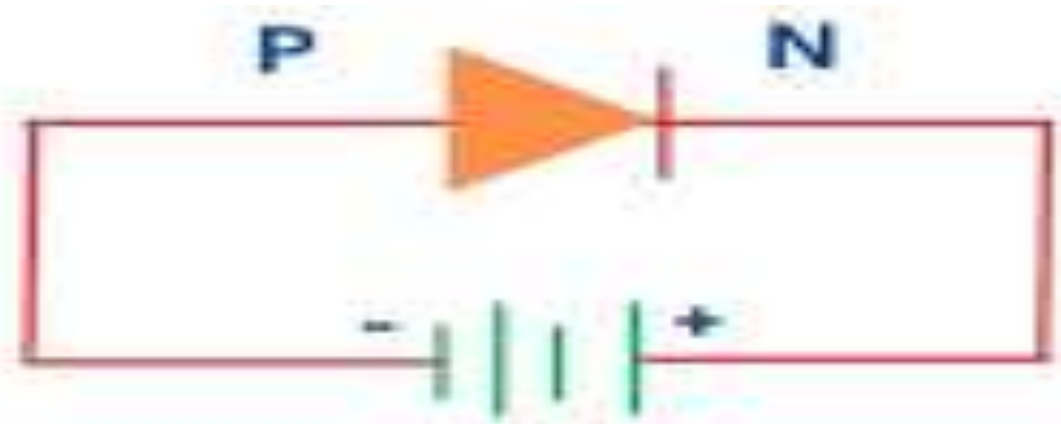
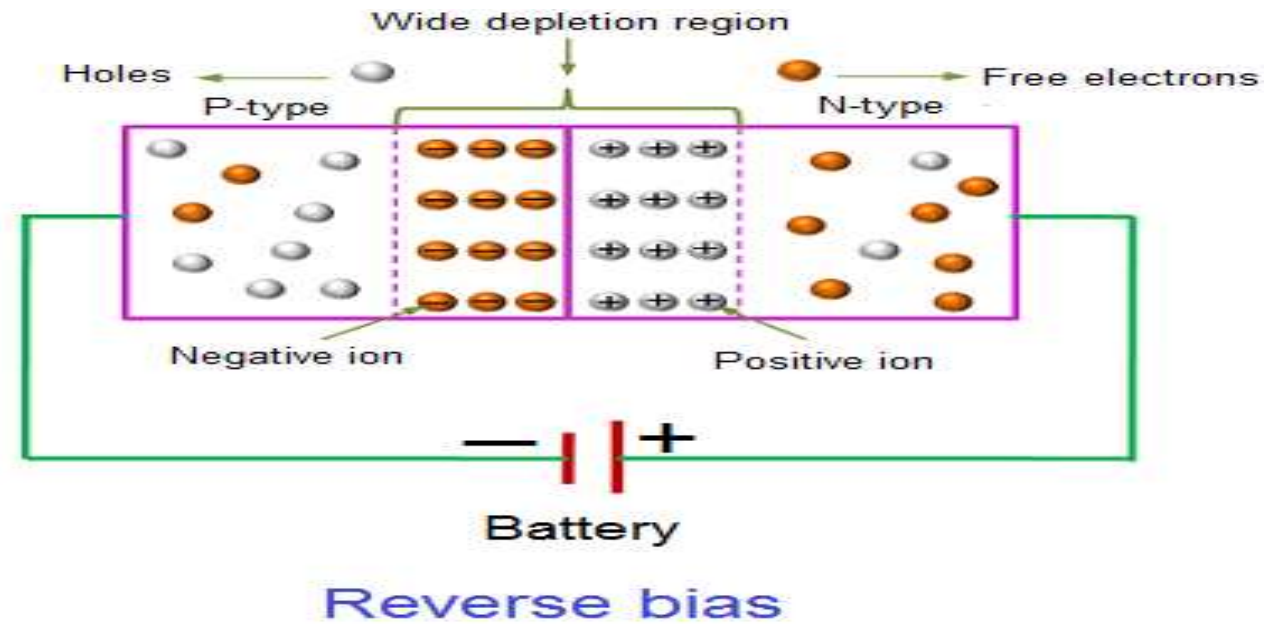
Electric Toothbrushes: Diodes regulate charging circuits to protect the battery.

Symbol	Name	Physical Look	Applications
	Diode		Mostly used as rectifier for converting ac to dc
	Zener Diode		Used as a constant voltage source
	Schottky Diode		High current diode used for high current
	Tunnel Diode		Used in oscillator and microwave amplifier
	Light Emitting Diode (LED)		Emit light when it is forward biased
	Photodiode		Allows reverse current when light falls on it
	Laser Diode		Used in CD players
	Current Regulator Diode		Maintains constant current
	Varactor Diode		Acts as variable capacitor in reverse bias
	Pin diode		Acts as capacitor in reverse and variable resistor in forward

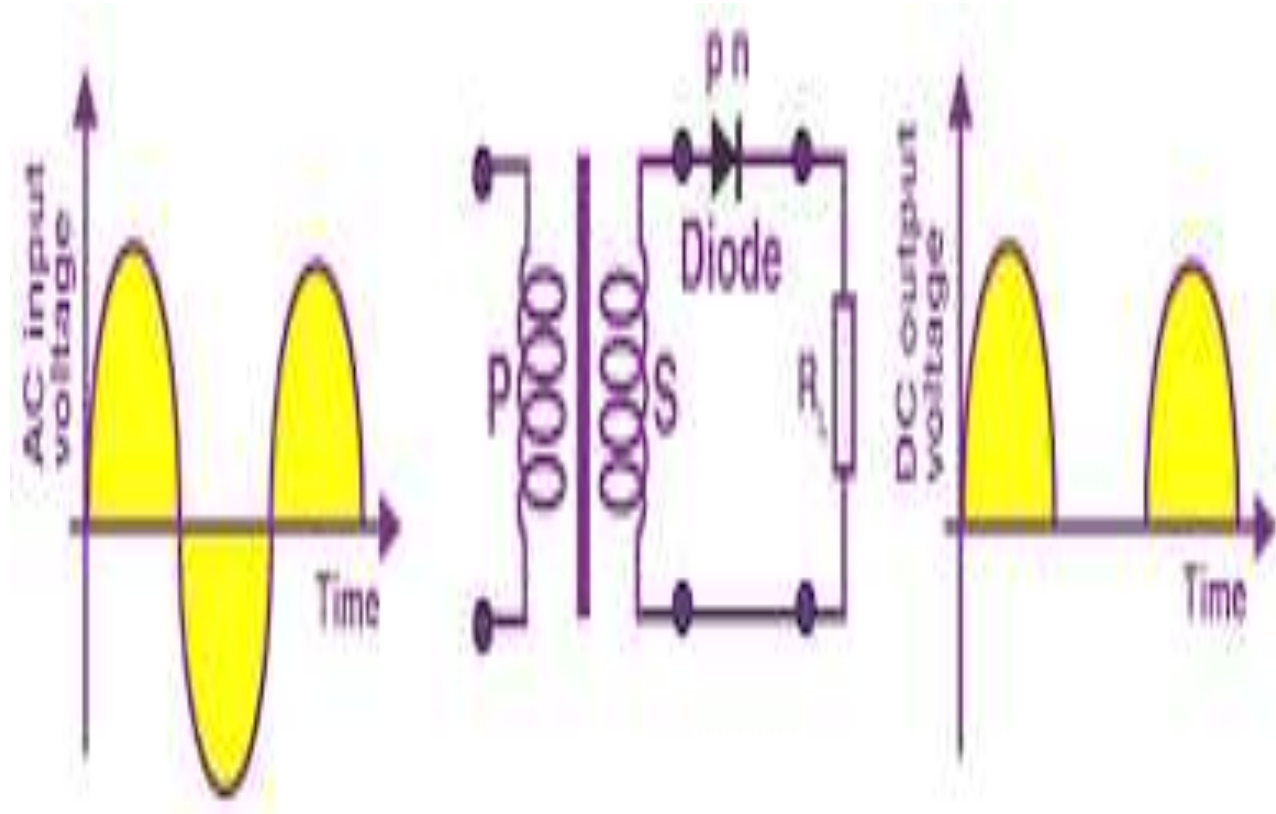
❖ **Forward biasing** a diode occurs when a voltage is applied, connecting the P-type region to the positive terminal and the N-type region to the negative terminal. This connection reduces the diode's internal potential barrier, making the depletion region thinner and allowing current to flow easily through the device. For a silicon diode, significant current flows once the forward voltage is around 0.7V.



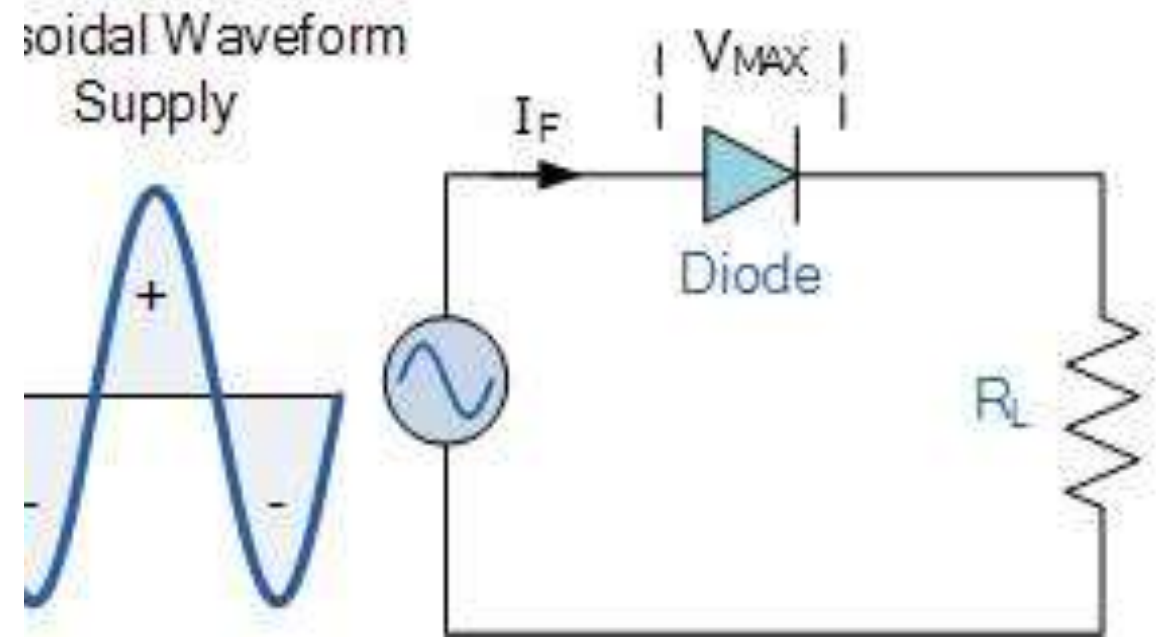
Reverse biasing a diode occurs when a voltage is applied, connecting the p-type material to the negative terminal and the n-type material to the positive terminal. This action increases the width of the depletion region, which widens the potential barrier and blocks the flow of current, allowing only a very small reverse leakage current to pass through. A large enough reverse voltage, known as the breakdown voltage, can cause a sudden and large increase in current that may damage the diode.



- ❖ A diode acts as a **rectifier** by converting **alternating current (AC) into direct current (DC)**, a process known as rectification. This is possible because a diode allows current to flow in only one direction (from anode to cathode) and blocks it in the reverse direction.
- ❖ When an AC voltage is applied across a diode, the diode becomes forward-biased during one half-cycle, allowing current to pass with very low resistance. During the other half-cycle, the diode is reverse-biased, presenting high resistance and effectively blocking the current flow. This results in a pulsating, unidirectional current at the output.



❖ **Half-Wave Rectifier:** This simple circuit uses a single diode and allows only one half of the AC waveform to pass through to the load. The result is a pulsating DC with significant "ripple" (remaining AC component). It is less efficient (about 40.6%) and typically used in low-power applications.



Half Wave Rectifier

